LESSON PLAN

AUXILIARY EQUIPMENT (22)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	5	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Description & operation of simple float type carburettor, modified carburettor, limitation Types of Engine starters, & trouble shooting on piston engine
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Description & operation of simple float type carburettor, modified carburettor, limitation Types of Engine starters, & trouble shooting on piston engine
5. Specific Objectives	22.1.1.1 Trainees will be able to explain about the working principle of piston engines.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of carburettor operation using cutaway model; Safety: Ensure model is disconnected from fuel source; Supervision: Instructor monitors trainee handling of components
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a carburettor in a piston engine?
(b) Name two types of engine starters used in aviation.
(c) List three common causes of carburettor icing.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Working principle of simple float type carburettor • Air velocity and pressure drop mechanism • Fuel metering and float level control • Choke operation and mixture enrichment	L & Ds	PPT, IB	15 min
(b)	• Modified carburettor features and improvements • Limitations of float type carburettors • Types of engine starters: Pneumatic and Electric • Troubleshooting carburettor issues and starter faults	L & Ds	PPT, PDF	20 min
(c)	• Detailed analysis of starter limitations • Common troubleshooting scenarios for carburettors • Preventive maintenance checks for starting systems • Review of engine starting sequence procedures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a modified carburettor overcomes the limitations of a simple float type.
(b) Describe the operational differences between pneumatic and electric engine starters.
(c) Identify two critical troubleshooting steps for a non-starting piston engine.

12. Home Assignment and Follow Up	Prepare a brief note comparing the fuel delivery mechanisms of a simple float carburettor and a modified carburettor, referencing the principles covered in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min